# Temporal bone & skull base

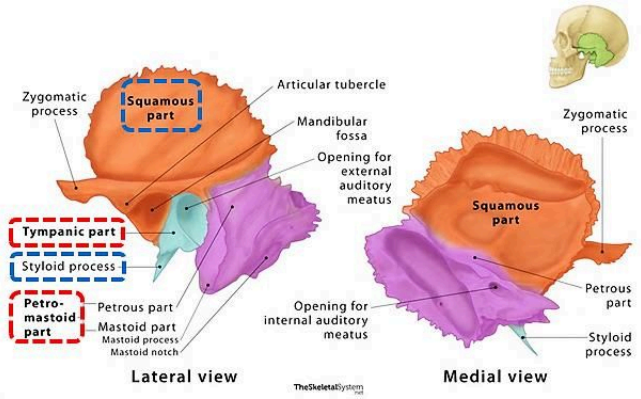
The bony house of the ear: five parts, three petrous surfaces, the foramina of the skull base and the landmarks you drill by.


1. Overview and contents
2. Petrous part: three surfaces
3. Tympanic and mastoid parts, styloid process
4. Foramina of the skull base
5. Pterygopalatine fossa
6. On imaging
7. In theatre


## 01 Overview and contents

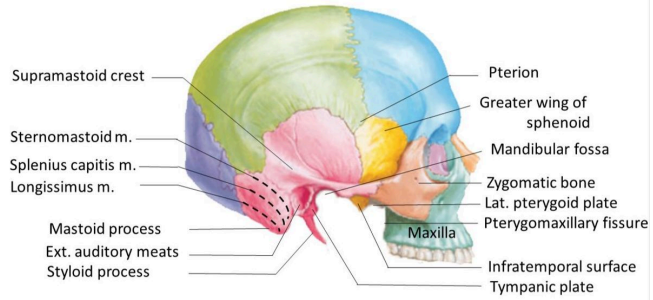
- Forms part of the **base and lateral wall** of the skull; its petrous part is pyramid-shaped. Articulates with 5 bones: occipital, parietal, sphenoid, zygomatic and the mandible (TMJ).
- The **petrous part separates the middle from the posterior cranial fossa**.
- **Houses:** bony EAC, middle ear with the ossicles, inner ear, the fallopian canal (facial nerve), the carotid canal (ICA), and bony cover of the sigmoid sinus and jugular bulb.
- **Five parts:** squamous, petrous, mastoid, tympanic and the styloid process.
- **Squamous part:** lateral wall of the middle cranial fossa and the **roof of the bony EAC**; its zygomatic process roofs the mandibular fossa.

**CLINICAL PEARL** The ear sits next to the middle and posterior fossae, the jugular bulb, sigmoid sinus, ICA and cranial nerves V–XII. This is why otitis media can produce intracranial complications.



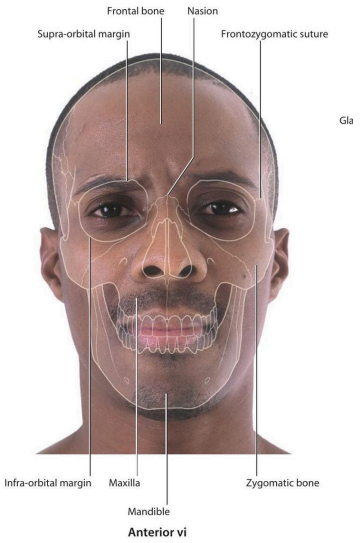

Parts of the temporal bone: lateral and medial views

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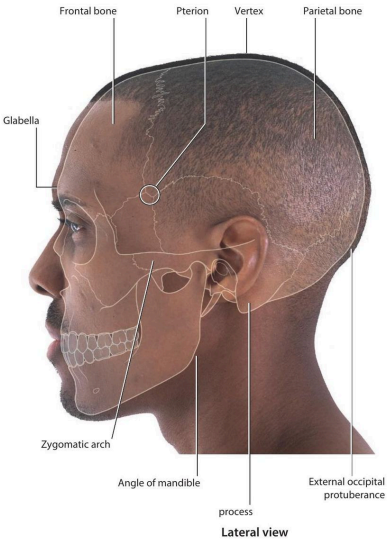

Lateral view of the skull: temporal and infratemporal regions

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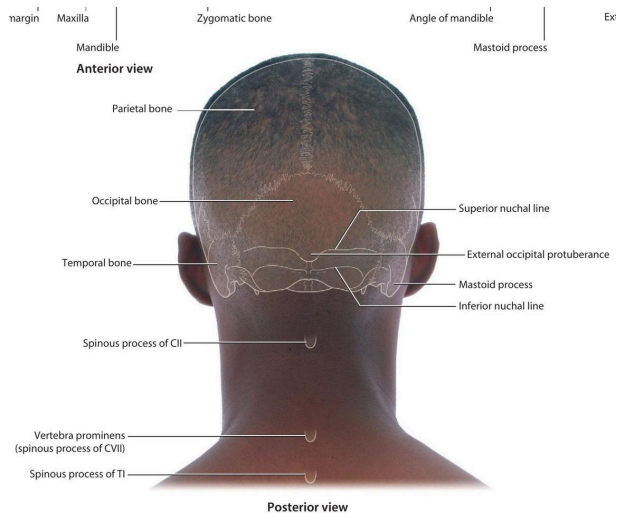

Surface anatomy with bones: anterior view

Gray's Atlas of Anatomy, 2nd ed., p. 476



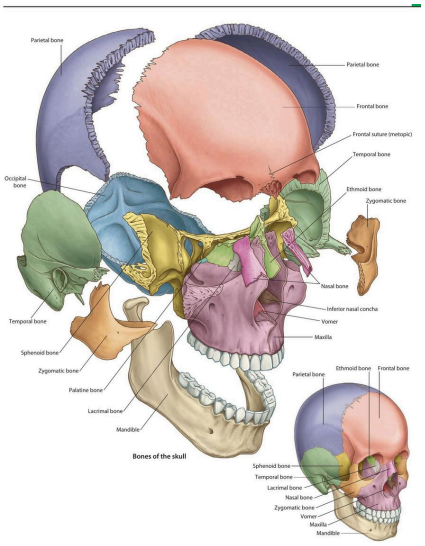

Surface anatomy with bones: lateral view (pterion, mastoid, angle)

Gray's Atlas of Anatomy, 2nd ed., p. 476




Surface anatomy with bones: posterior view

Gray's Atlas of Anatomy, 2nd ed., p. 476



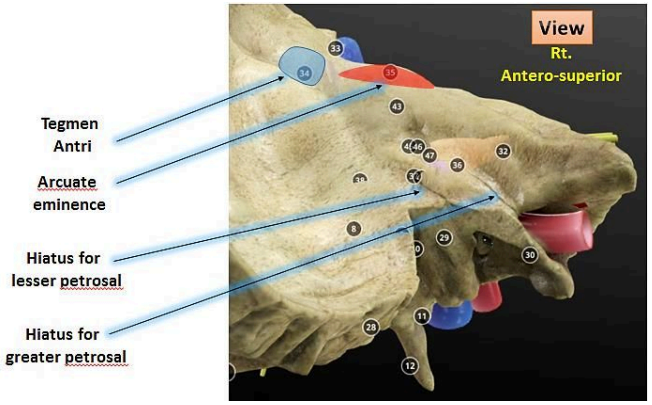

Bones of the skull (exploded view)

Gray's Atlas of Anatomy, 2nd ed., p. 477

## 02 Petrous part: three surfaces

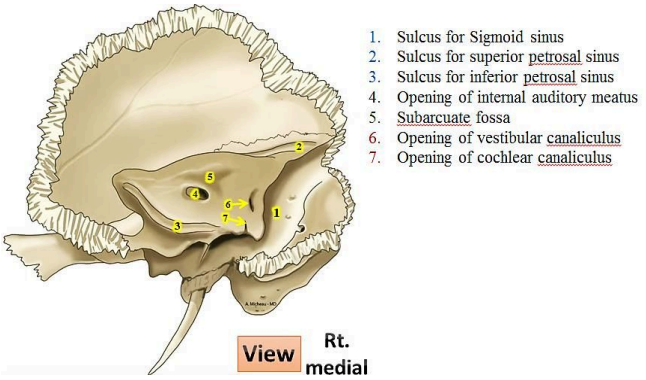
- **Anterior surface** (floor of the middle cranial fossa): **arcuate eminence** of the superior canal medially, **tegmen tympani and tegmen antri** laterally, canal for tensor tympani and the ET in front, **trigeminal impression** at the apex, and hiatuses for the greater and lesser petrosal nerves.
- **Posterior surface** (front of the posterior fossa): **internal acoustic meatus** (VII, VIII, labyrinthine vessels), **subarcuate fossa** superolateral to the IAM, the **operculum** for the vestibular aqueduct inferolateral to it, and grooves for the sigmoid, superior and inferior petrosal sinuses.
- **Inferior surface:** jugular fossa (jugular bulb), lower opening of the **carotid canal** in front of it, separated by a wedge of bone called the **keel (caroticojugular crest)**.
- The **tympanic canaliculus** in the keel transmits **Jacobson's nerve** (tympanic branch of IX). The **cochlear aqueduct** opens at the top of the jugular foramen.
- The **jugular foramen** forms where the petrous and occipital bones meet.

**CLINICAL PEARL** Petrous apicitis (Gradenigo's syndrome): ear discharge, retro-orbital pain (V at the apex) and lateral rectus palsy (VI in Dorello's canal).



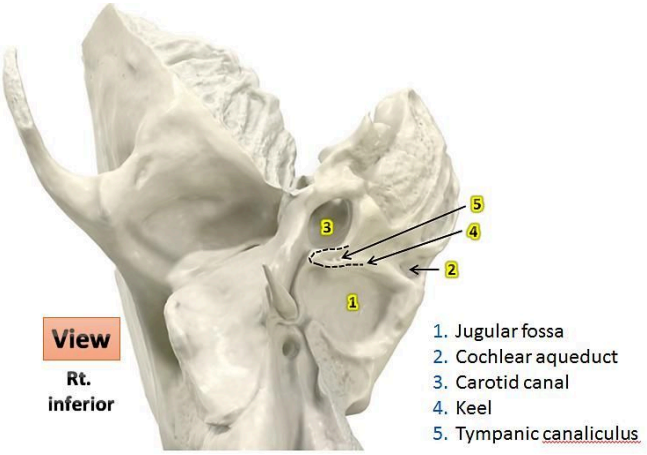

Anterior surface of the petrous bone: tegmen, arcuate eminence, petrosal hiatuses

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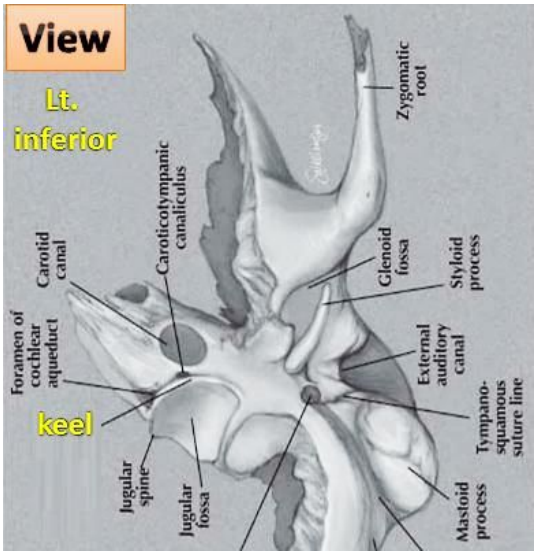

Posterior surface of the petrous bone: IAM, subarcuate fossa, sinus sulci

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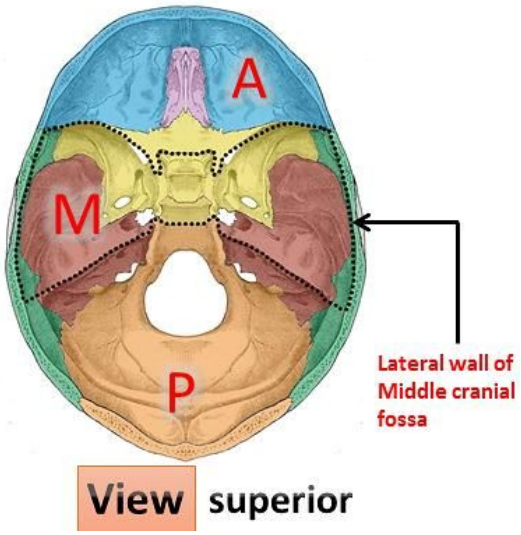

Inferior surface: jugular fossa, carotid canal, cochlear aqueduct

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Inferior surface in detail with the keel between carotid and jugular

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Cranial fossae from above: the petrous ridge divides middle from posterior fossa

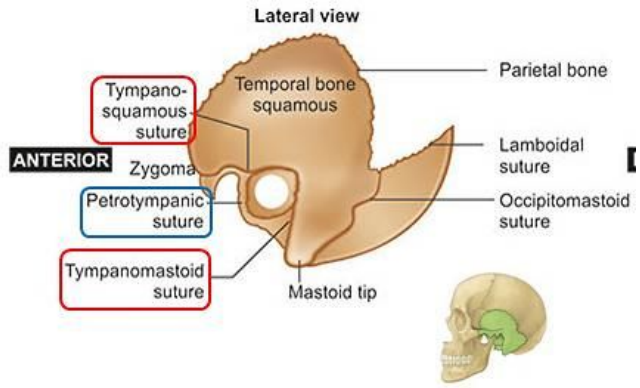
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## 03 Tympanic and mastoid parts, styloid process

- **Tympanic part:** forms the anterior wall, floor and posterior wall of the bony EAC, the anterior wall and floor of the middle ear, and the back of the mandibular fossa.
- **Sutures:** in the bony EAC, **tympanosquamous** in front and **tympanomastoid** behind; in the middle ear, the **petrotympanic (Glaserian) fissure**, through which the chorda tympani leaves.
- **Mastoid part:** the tip receives **SCM**; the **digastric groove** gives origin to the posterior belly of digastric and ends in front at the **stylomastoid foramen**; an emissary vein to the sigmoid sinus passes through the mastoid foramen.
- The **mastoid antrum lies deep to Macewen's (suprameatal) triangle**.
- **Styloid process:** in front of the stylomastoid foramen. **Three muscles** (stylohyoid, styloglossus, stylopharyngeus) and **two ligaments** (stylohyoid, stylomandibular).

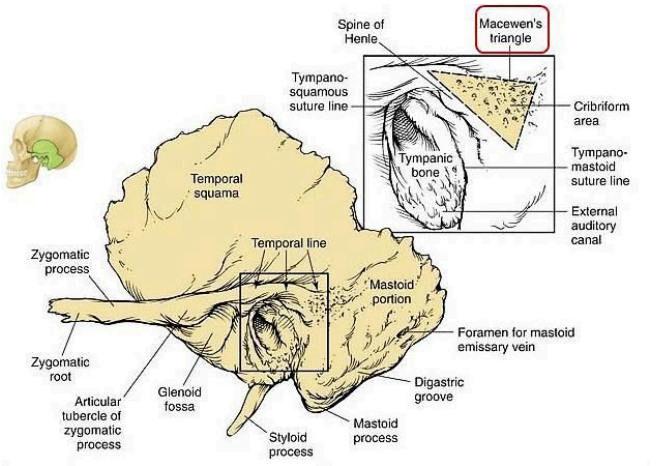
**CLINICAL PEARL** Stylomandibular ligament separates the parotid from the submandibular gland; an elongated styloid (over 3 cm) causes Eagle's syndrome.

**SOURCES DIFFER CHORDA TYMPANI EXIT:** ENT Steps says the petrotympanic fissure; Doss says the squamotympanic fissure. Gray's: the anterior canaliculus (canal of Huguier) at the medial end of the petrotympanic fissure.



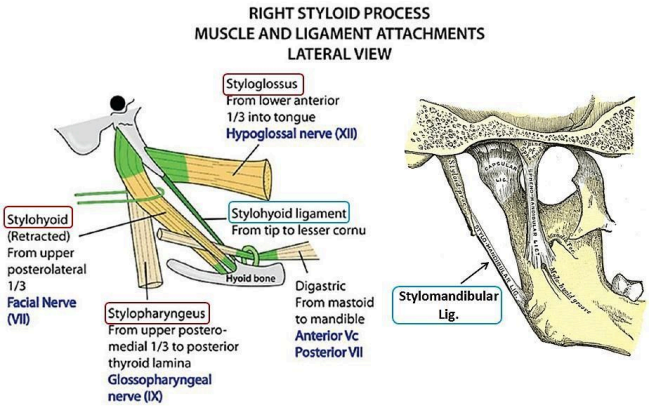

Sutures of the temporal bone, lateral view

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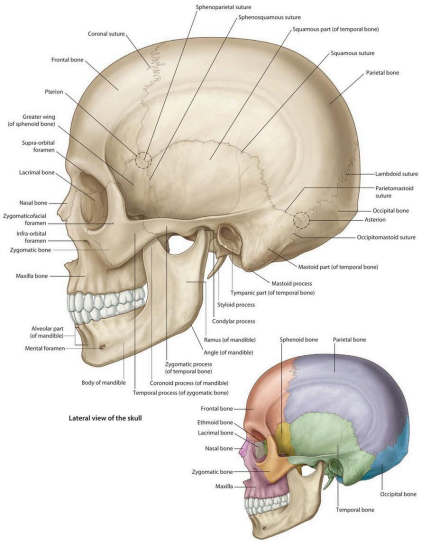

Lateral surface: spine of Henle, Macewen's triangle, mastoid

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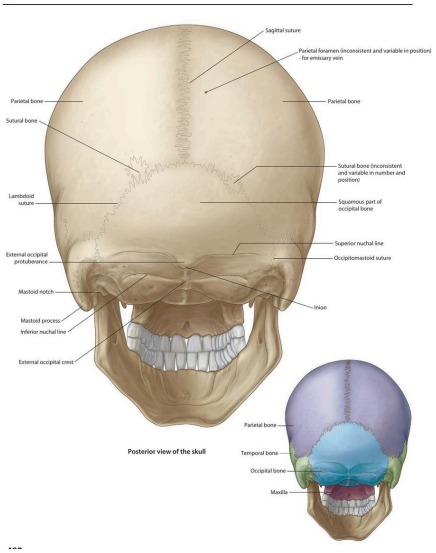

Styloid apparatus: 3 muscles and 2 ligaments

ENT Steps Anatomy, p. 27




Lateral view of the skull: squamous, tympanic and mastoid parts

Gray's Atlas of Anatomy, 2nd ed., p. 480




Posterior view of the skull: mastoid process and notch, occipital bone

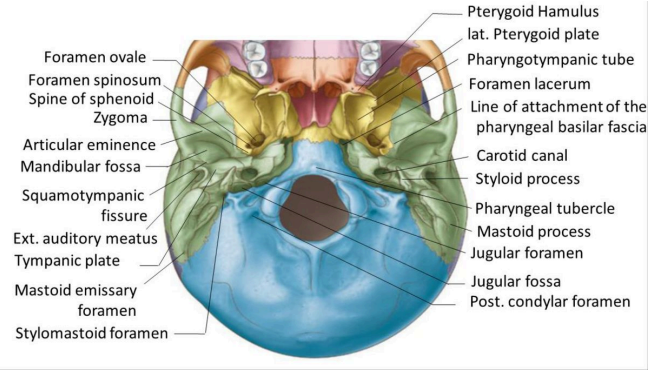
Gray's Atlas of Anatomy, 2nd ed., p. 482

## 04 Foramina of the skull base

- **Cribriform plate:** olfactory nerves (anterior fossa).
- **Optic canal:** optic nerve and ophthalmic artery.
- **Superior orbital fissure:** III, IV, VI, the three branches of V1 (lacrimal, frontal, nasociliary) and the superior ophthalmic vein.
- **Foramen rotundum:** V2. **Foramen ovale (MALE):** Mandibular nerve, Accessory meningeal artery, Lesser petrosal nerve, Emissary vein. **Foramen spinosum:** middle meningeal artery and the meningeal branch of V3.
- **Foramen lacerum:** filled with cartilage; the ICA crosses its upper part, and the greater petrosal joins the deep petrosal to form the vidian nerve.
- **Internal acoustic meatus:** VII, nervus intermedius, VIII and labyrinthine vessels.
- **Jugular foramen:** IX, X, XI, the IJV (from the sigmoid sinus) and the inferior petrosal sinus. **Hypoglossal canal:** XII. **Stylomastoid foramen:** VII and the stylomastoid artery.

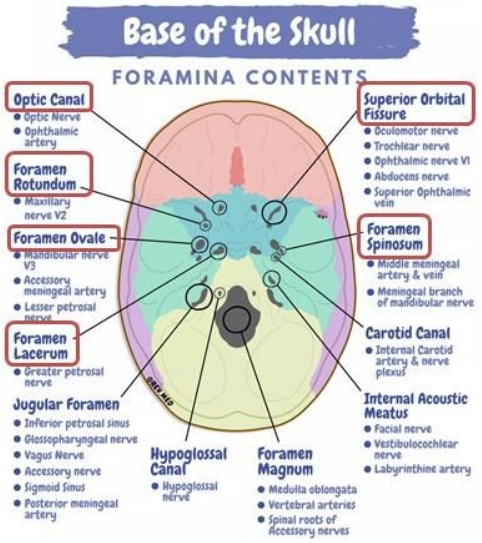
**CLINICAL PEARL** Jugular foramen syndromes: Vernet = IX, X, XI. Collet–Sicard = IX–XII. Villaret = IX–XII plus the cervical sympathetic chain (Horner's).

**SOURCES DIFFER FORAMEN OVALE:** Doss adds the motor root of V and puts the middle meningeal vein here ("3 M + ALE"); Gray's sends the middle meningeal vein through the foramen spinosum.



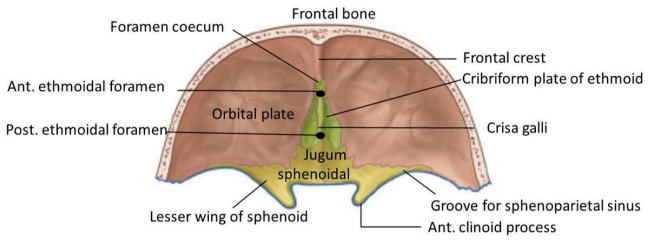

Norma basalis: foramina of the skull base

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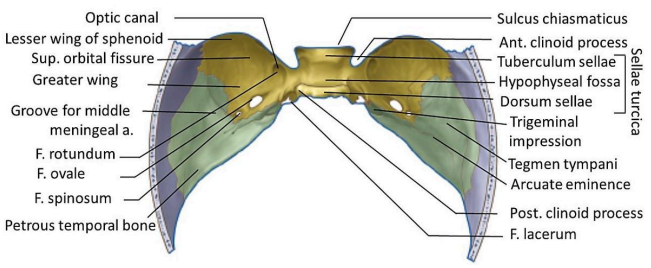

Foramina of the skull base and their contents

ENT Steps Anatomy, p. 152



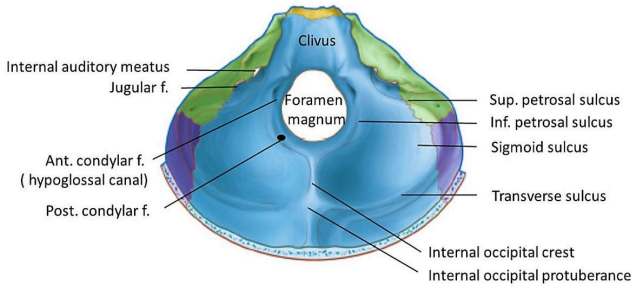

Anterior cranial fossa

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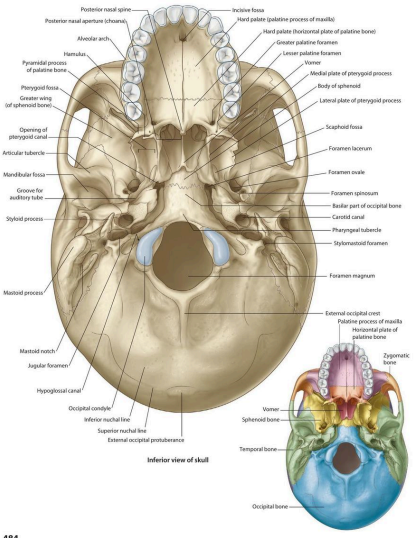

Middle cranial fossa and its foramina

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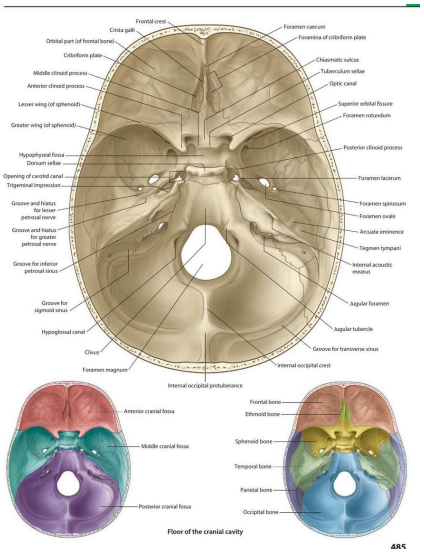

Posterior cranial fossa

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Inferior view of the skull: foramina of the skull base

Gray's Atlas of Anatomy, 2nd ed., p. 484

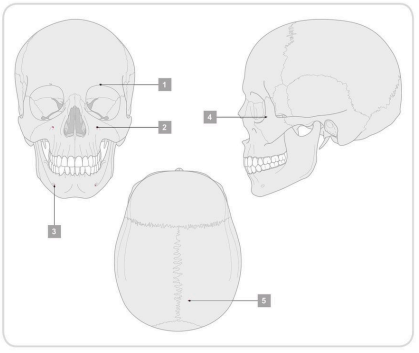



Floor of the cranial cavity: cranial fossae and foramina

Gray's Atlas of Anatomy, 2nd ed., p. 485

*External foramina of the skull*

| Foramen | | Structures passing through foramen |
|---|---|---|
| *Anterior view* | | |
| Supra-orbital foramen | 1 | Supra-orbital nerve and vessels |
| Infra-orbital foramen | 2 | Infra-orbital nerve and vessels |
| Mental foramen | 3 | Mental nerve and vessels |
| *Lateral view* | | |
| Zygomaticofacial foramen | 4 | Zygomaticofacial nerve |
| *Superior view* | | |
| Parietal foramen | 5 | Emissary veins |



External foramina of the skull and their contents

Gray's Atlas of Anatomy, 2nd ed., p. 587

*External foramina of the skull*

| Foramen | | Structures passing through foramen |
|---|---|---|
| *Inferior view* | | |
| Incisive foramina | 1 | Nasopalatine nerve; sphenopalatine vessels |
| Greater palatine foramen | 2 | Greater palatine nerve and vessels |
| Lesser palatine foramina | 3 | Lesser palatine nerves and vessels |
| Pterygoid canal | 4 | Pterygoid nerve and vessels |
| Foramen ovale | 5 | Mandibular nerve [$V_3$]; lesser petrosal nerve |
| Foramen spinosum | 6 | Middle meningeal artery |
| Foramen lacerum | 7 | Filled with cartilage |
| Carotid canal | 8 | Internal carotid artery and nerve plexus |
| Foramen magnum | 9 | Continuation of brain and spinal cord; vertebral arteries and nerve plexuses; anterior spinal artery; posterior spinal arteries; roots of accessory nerve [XI]; meninges |
| Condylar canal | 10 | Emissary veins |
| Hypoglossal canal | 11 | Hypoglossal nerve [XII] and vessels |
| Jugular foramen | 12 | Internal jugular vein; inferior petrosal sinus; glossopharyngeal nerve [IX]; vagus nerve [X]; accessory nerve [XI] |
| Stylomastoid foramen | 13 | Facial nerve [VII] |

*Internal foramina of the skull*

| | | |
|---|---|---|
| *Anterior cranial fossa* | | |
| Foramen caecum | 1 | Emissary veins to nasal cavity |
| Olfactory foramina in cribriform plate | 2 | Olfactory nerves [I] |
| *Middle cranial fossa* | | |
| Optic canal | 3 | Optic nerve [II]; ophthalmic artery |
| Superior orbital fissure | 4 | Oculomotor nerve [III]; trochlear nerve [IV]; ophthalmic division of the trigeminal nerve [$V_1$]; abducent nerve [VI]; ophthalmic veins |
| Foramen rotundum | 5 | Maxillary division of the trigeminal nerve [$V_2$] |
| Foramen ovale | 6 | Mandibular division of the trigeminal nerve [$V_3$]; lesser petrosal nerve |
| Foramen spinosum | 7 | Middle meningeal artery |
| Hiatus for the greater petrosal nerve | 8 | Greater petrosal nerve |
| Hiatus for the lesser petrosal nerve | 9 | Lesser petrosal nerve |
| *Posterior cranial fossa* | | |
| Foramen magnum | 10 | End of brainstem/beginning of spinal cord; vertebral arteries; spinal roots of the accessory nerve; meninges |
| Internal acoustic meatus | 11 | Facial nerve [VII]; vestibulocochlear nerve [VIII]; labyrinthine artery |
| Jugular foramen | 12 | Glossopharyngeal nerve [IX]; vagus nerve [X]; accessory nerve [XI]; inferior petrosal sinus, sigmoid sinus (forming internal jugular vein) |
| Hypoglossal canal | 13 | Hypoglossal nerve [XII]; meningeal branch of the ascending pharyngeal artery |
| Condylar canal | 14 | Emissary vein |

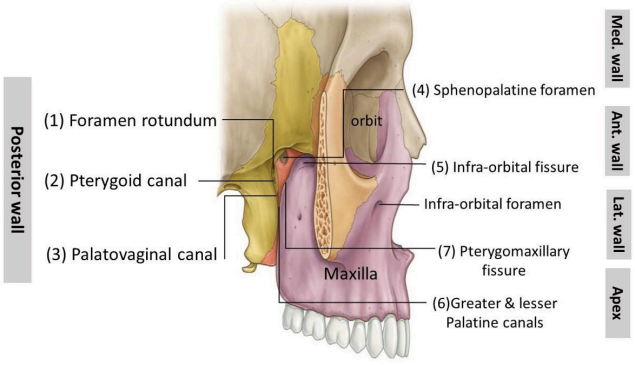
Internal foramina of the skull and their contents

Gray's Atlas of Anatomy, 2nd ed., p. 588

## 05 Pterygopalatine fossa

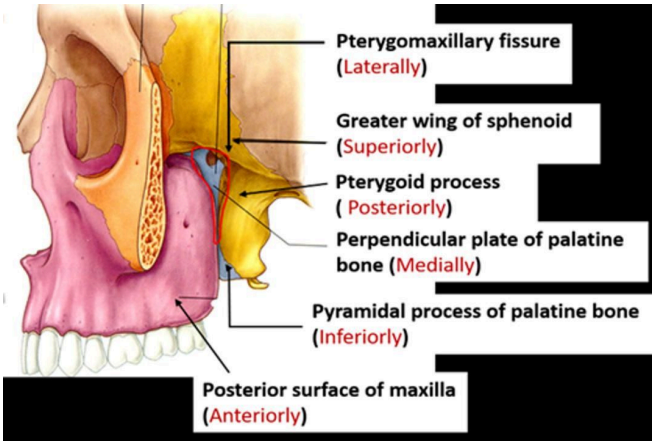
- Small inverted pyramidal space behind the maxilla. **Roof:** body of sphenoid. **Front:** maxilla. **Back:** pterygoid plates. **Medial:** perpendicular plate of the palatine bone.
- **Contents:** third part of the maxillary artery, the maxillary nerve (V2) and the **pterygopalatine (sphenopalatine) ganglion**.
- **Communications:** orbit (inferior orbital fissure), middle cranial fossa (foramen rotundum), foramen lacerum (vidian canal), nasopharynx (palatovaginal canal), nasal cavity (**sphenopalatine foramen**), infratemporal fossa (**pterygomaxillary fissure**), oral cavity (greater and lesser palatine canals).

**CLINICAL PEARL** Juvenile angiofibroma starts at the sphenopalatine foramen and spreads through these communications. Widening of the pterygopalatine fossa pushes the back wall of the maxilla forward (Holman–Miller sign).



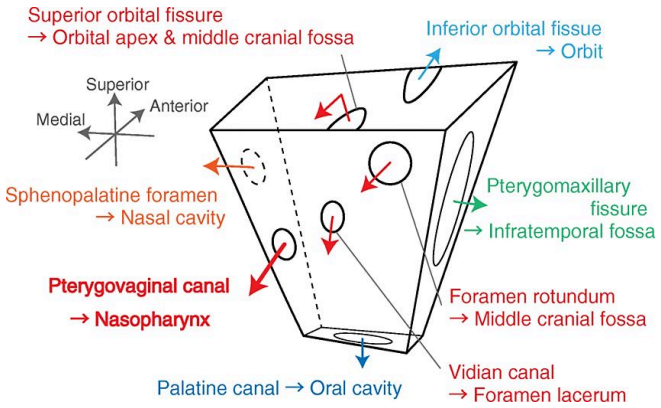

Pterygopalatine fossa and its openings

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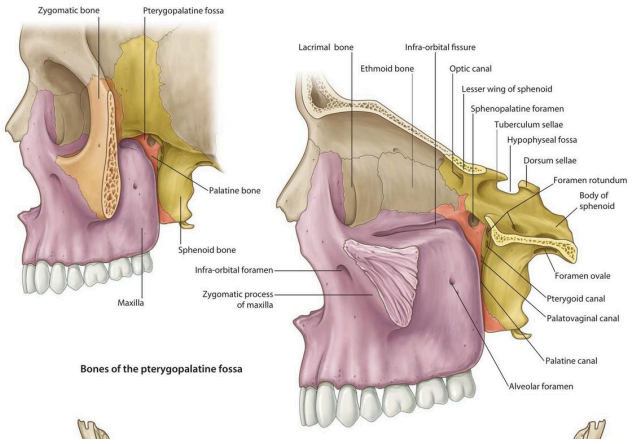

Boundaries of the pterygopalatine fossa

ENT Steps Anatomy, p. 160




Communications of the pterygopalatine fossa

ENT Steps Anatomy, p. 161




Bones of the pterygopalatine fossa

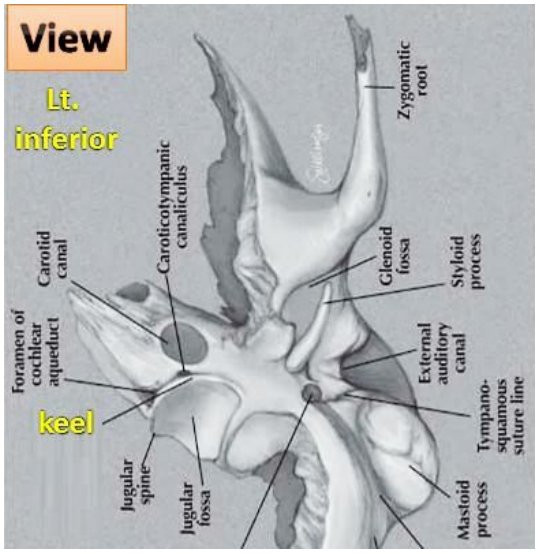
Gray's Atlas of Anatomy, 2nd ed., p. 534

## 06 Radiological anatomy: On imaging

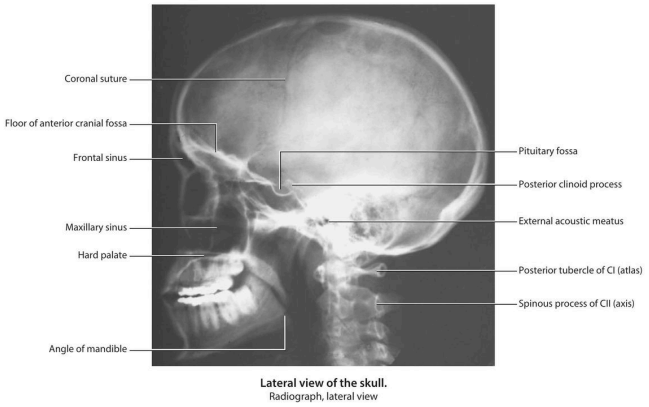
- **HRCT temporal bone** (axial and coronal, about 0.6 mm slices, bone window) is the standard for bony anatomy; **MRI with gadolinium** (T1, T2/CISS) shows the IAC, cochlear fluid, nerves and soft tissue.
- On axial CT, find in order from top: superior canal, then the IAC with the vestibule and lateral canal, then the cochlea and the tympanic facial nerve, then the jugular bulb and carotid canal.

- **Look for:** tegmen thickness and dehiscence, sigmoid sinus position, high or dehiscent jugular bulb, carotid canal dehiscence, superior canal dehiscence (Pöschl view).

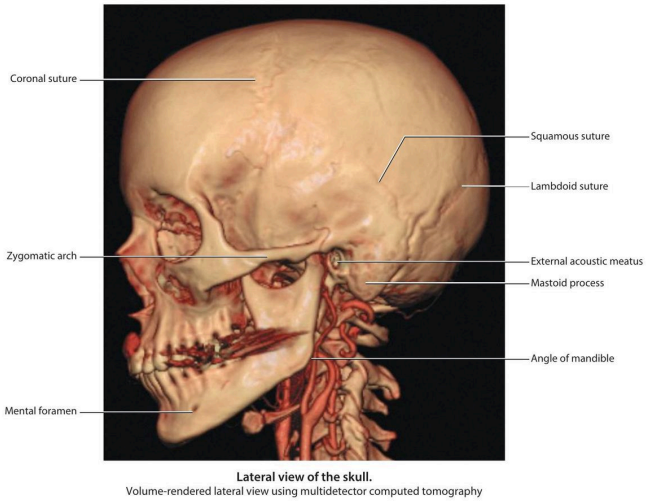
> **CLINICAL PEARL** A high-riding jugular bulb above the round window level can be injured during cochlear implantation or myringotomy.




Inferior surface in detail with the keel between carotid and jugular
ENT Steps Anatomy, p. 24




Lateral skull radiograph: pituitary fossa, sinuses, external acoustic meatus
Gray's Atlas of Anatomy, 2nd ed., p. 481




Lateral view of the skull: volume-rendered CT
Gray's Atlas of Anatomy, 2nd ed., p. 481

## 07 Surgical anatomy: In theatre

- Skull base approaches are planned around the **ICA, jugular bulb, facial nerve, labyrinth and dura** seen on CT.
- The **arcuate eminence** marks the superior canal only roughly; in the middle fossa approach the greater petrosal nerve and the IAC are safer landmarks.
- **Infratemporal fossa approach (Fisch A)** permanently reroutes the facial nerve anteriorly to reach the jugular foramen.

# Tables

### Skull base foramina and their contents

| Foramen | Bone / fossa | Contents |
|---|---|---|
| **Cribriform plate** | Ethmoid · anterior fossa | Olfactory nerves (I), anterior ethmoidal nerve and vessels |
| **Optic canal** | Lesser wing of sphenoid | Optic nerve (II), ophthalmic artery |
| **Superior orbital fissure** | Between greater and lesser wings | III, IV, VI, V1 (lacrimal, frontal, nasociliary), superior ophthalmic vein |
| **Foramen rotundum** | Greater wing · middle fossa | Maxillary nerve (V2) |
| **Foramen ovale** | Greater wing · middle fossa | V3, accessory meningeal artery, lesser petrosal nerve, emissary vein (MALE) |
| **Foramen spinosum** | Greater wing · middle fossa | Middle meningeal artery and vein, meningeal branch of V3 |
| **Foramen lacerum** | Between sphenoid, petrous and occipital | Cartilage; ICA crosses its upper part; greater + deep petrosal form the vidian nerve |
| **Internal acoustic meatus** | Petrous · posterior fossa | VII, nervus intermedius, VIII, labyrinthine artery and vein |
| **Jugular foramen** | Petrous + occipital · posterior fossa | IX, X, XI, IJV (sigmoid sinus), inferior petrosal sinus |
| **Hypoglossal canal** | Occipital | XII, meningeal branch of ascending pharyngeal artery |
| **Stylomastoid foramen** | Temporal | Facial nerve (VII), stylomastoid artery |
| **Petrotympanic fissure** | Temporal | Chorda tympani, anterior tympanic artery |
| **Foramen magnum** | Occipital | Medulla, vertebral arteries, spinal roots of XI |

Jugular foramen syndromes: Vernet IX–XI, Collet–Sicard IX–XII, Villaret IX–XII + sympathetic chain.

### Pharyngeal arch derivatives

| Arch | Nerve | Skeletal | Muscles |
|---|---|---|---|
| **1st** | V3 | Malleus (head), incus (body), Meckel's cartilage, sphenomandibular ligament | Mastication, tensor tympani, tensor veli palatini, mylohyoid, anterior digastric |
| **2nd** | VII | Stapes (superstructure), styloid, stylohyoid ligament, lesser horn and upper hyoid body | Facial expression, stapedius, stylohyoid, posterior digastric |
| **3rd** | IX | Greater horn and lower hyoid body | Stylopharyngeus |
| **4th** | X (superior laryngeal) | Thyroid cartilage | Cricothyroid, pharyngeal constrictors, levator veli palatini |
| **6th** | X (recurrent laryngeal) | Cricoid, arytenoid, corniculate | Intrinsic laryngeal muscles (except cricothyroid) |

Pouches: 1st → middle ear and ET; 2nd → tonsillar fossa; 3rd → inferior parathyroid and thymus; 4th → superior parathyroid and C cells.

### Cranial fossae relations

| Fossa | Floor formed by | ENT relations |
|---|---|---|
| **Anterior** | Orbital plates of frontal, cribriform plate, lesser wings | Frontal sinus, ethmoid roof, olfactory bulbs |

| Fossa | Floor formed by | ENT relations |
|---|---|---|
| **Middle** | Greater wings of sphenoid, squamous and petrous temporal | Tegmen, sphenoid sinus, cavernous sinus, foramina rotundum, ovale, spinosum |
| **Posterior** | Occipital bone, posterior petrous surface | IAM, jugular foramen, sigmoid sinus, cerebellum (mastoid relation) |

## M Mnemonics

Foramen ovale contents

**MALE**

Mandibular nerve, Accessory meningeal artery, Lesser petrosal nerve, Emissary vein

Superior orbital fissure

**Live Frankly To See No Insult At all**

Lacrimal, Frontal, Trochlear, Superior division of III, Nasociliary, Inferior division of III, Abducens (plus the superior ophthalmic vein)

## Q Rapid recall

1. **Name the five parts of the temporal bone.** Squamous, petrous, mastoid, tympanic and styloid process.
2. **What is the keel of the temporal bone and what passes through it?** The caroticojugular crest between the carotid canal and jugular fossa; the tympanic canaliculus in it carries Jacobson's nerve (IX).
3. **Contents of the foramen ovale?** MALE: Mandibular nerve, Accessory meningeal artery, Lesser petrosal nerve, Emissary vein.
4. **Contents of the jugular foramen?** IX, X, XI, the IJV (from the sigmoid sinus) and the inferior petrosal sinus.
5. **Triad of Gradenigo's syndrome?** Ear discharge, retro-orbital pain (V) and lateral rectus palsy (VI) from petrous apicitis.
6. **Which cranial nerves are involved in Vernet, Collet–Sicard and Villaret syndromes?** Vernet IX–XI; Collet–Sicard IX–XII; Villaret IX–XII plus cervical sympathetic (Horner's).
7. **Three muscles and two ligaments attached to the styloid process?** Stylohyoid, styloglossus, stylopharyngeus; stylohyoid and stylomandibular ligaments.
8. **Through which fissure does the chorda tympani leave the skull?** Petrotympanic (Glaserian) fissure.
9. **What lies medially on the anterior surface of the petrous bone?** The arcuate eminence, raised by the superior semicircular canal.
10. **Where is the trigeminal impression?** At the apex of the anterior petrous surface, for the trigeminal ganglion in Meckel's cave.
11. **Which petrous surface carries the internal acoustic meatus?** The posterior surface (front of the posterior cranial fossa).
12. **What is the subarcuate fossa?** Depression superolateral to the IAM, under the arcuate eminence, carrying the subarcuate artery.
13. **Where does the cochlear aqueduct open externally?** At the upper limit of the jugular foramen on the inferior petrous surface.
14. **Which part of the temporal bone forms most of the bony EAC?** The tympanic part (the squamous part forms its roof).
15. **Two sutures in the bony EAC?** Tympanosquamous in front, tympanomastoid behind.
16. **Where does the posterior belly of digastric arise?** Digastric groove (mastoid notch) on the medial side of the mastoid.
17. **Contents of the internal acoustic meatus?** Facial nerve, nervus intermedius, vestibulocochlear nerve, labyrinthine artery and vein.
18. **Boundaries of the pterygopalatine fossa?** Roof: sphenoid body. Front: maxilla. Back: pterygoid plates. Medial: perpendicular plate of palatine.

19. **What passes through the foramen lacerum in life?** Only small structures pass through its cartilage (meningeal branch of ascending pharyngeal artery, emissary veins); the ICA crosses its upper part and the vidian nerve forms there.
20. **What separates the middle from the posterior cranial fossa?** The petrous ridge of the temporal bone (with the tentorium attached).

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Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.